EVERY SUMMAND OF AN OBJECT WITH A LEFT DUAL HAS A LEFT DUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the opening sentence 'We will use Categories, Lemma categories-lemma-left-dual without further mention.' names the same technique applied at each step rather than declining a step, and every identification the argument rests on is displayed or stated inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Left duals and the adjunction with the tensor product. Categories, Lemma categories-lemma-left-dual and Remark categories-remark-left-dual-adjoint. Supply the identification of Hom(X tensor Y, Z) with Hom(X, Z tensor Y) and the criterion for an object to be a left dual of another, which the proof uses throughout.
- Karoubian categories and split idempotents. Same chapter, Section section-karoubian. Supplies that an idempotent endomorphism of an object produces a direct sum decomposition, applied to the idempotents obtained in the endomorphism algebra of the dual.

Source reference. Original lemma 0FFQ: sources/stacks-categories/pinned-original-source.tex, source lines 9558–9558. Navigation pointer invoked at proof line 4223; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 09SF: sources/stacks-homology/pinned-original-source.tex, source lines 725–725. Navigation pointer invoked at proof line 4240;

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 0FFR: sources/stacks-categories/pinned-original-source.tex, source lines 9603–9603. Navigation pointer invoked at proof line 4213; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *In a Karoubian additive monoidal category every summand of an object which has a left dual has a left dual.*

Proof. We will use Categories, Lemma 0FFQ without further mention. Let X be an object which has a left dual Y. We have

$$\mathrm{Hom}(X, X) = \mathrm{Hom}(\mathbf{1}, X \otimes Y) = \mathrm{Hom}(Y, Y)$$

If $a : X \to X$ corresponds to $b : Y \to Y$ then b is the unique endomorphism of Y such that precomposing by a on

$$\mathrm{Hom}(Z' \otimes X, Z) = \mathrm{Hom}(Z', Z \otimes Y)$$

is the same as postcomposing by $1 \otimes b$. Hence the bijection $\mathrm{Hom}(X, X) \to \mathrm{Hom}(Y, Y)$, $a \mapsto b$ is an isomorphism of the opposite of the algebra $\mathrm{Hom}(X, X)$ with the algebra $\mathrm{Hom}(Y, Y)$. In particular, if $X = X_1 \oplus X_2$, then the corresponding projectors e_1, e_2 are mapped to idempotents in $\mathrm{Hom}(Y, Y)$. If $Y = Y_1 \oplus Y_2$ is the corresponding direct sum decomposition of Y (Section 09SF) then we see that under the bijection $\mathrm{Hom}(Z' \otimes X, Z) = \mathrm{Hom}(Z', Z \otimes Y)$ we have $\mathrm{Hom}(Z' \otimes X_i, Z) = \mathrm{Hom}(Z', Z \otimes Y_i)$ functorially as subgroups for $i = 1, 2$. It follows that Y_i is the left dual of X_i by the discussion in Categories, Remark 0FFR. $\square$